

# Communicating with Data – Perception

Dr. Ab Mosca (they/them)

Slides based off slides courtesy of Jordan Crouser (<https://jcrouser.github.io/>)

Sponsored by the Center for the Environment, Ecological Design and Sustainability, Environmental Science and Policy Program, Smith Office for the Arts, Lazarus Center for Career Development, Program in Statistical & Data Sciences, Sustainable Capital Programs, Endowed Lecture Fund, and the departments of Computer Science, Mathematical Sciences, and East Asian Languages and Cultures.

## From Computer Science to Center Stage

### A CONVERSATION WITH VIENNA TENG

Acclaimed singer-songwriter, performing artist, and creative entrepreneur Vienna Teng will lead an inspiring conversation about her journey from studying computer science and working at Cisco Systems, a leading technology company, to pursuing a full-time music career.



MONDAY, SEPT. 21

12-1:30 P.M.

McCartney Hall 106

REGISTER



THE SMITH OFFICE FOR THE ARTS PRESENTS:

## MUSIC FOR DESSERT:

SONGS & CHAT  
WITH  
VIENNA TENG!

SEPT 21 - 7:30PM  
ALUMNAE GYM 3<sup>RD</sup> FLOOR

VIENNA TENG IS A SONGWRITER, PERFORMER AND ORGANIZER. SHE HAS BEEN FEATURED ON NPR AND THE LATE SHOW, RECORDED ALBUMS FOR ROUNDER/UNIVERSAL RECORDS, AND HER MUSIC GARNERS OVER 450,000 ANNUAL LISTENERS ON STREAMING SERVICES.

- HEAR HER PLAY A FEW SONGS (POSSIBLY JOIN IN)
- CHAT ABOUT HER MUSIC CAREER
- EAT DESSERT

SmithArts



## START WITH JOY

### A CLIMATE ACTION WORKSHOP with Vienna Teng

Join Vienna Teng, acclaimed singer-songwriter, former McKinsey Sustainability fellow and one of the Grist 50 "climate innovators to watch," for the afternoon!

Rather than focusing on the climate crisis itself, Vienna will help you reflect on what gives you energy and meaning (start with joy), identify whom you could join in taking action (stop being one person), integrate climate solutions into your busy life (stack with other stuff), and choose one thing and take a simple first step (stress less). Come play with sticky notes, enjoy delicious food, hear a little music - and take your place in the epic, unfinished story of climate action.



MONDAY, SEPT. 21

4:30-6:30 P.M.

CEEDS, Wright Hall 005

REGISTER



CEEDS  
Center for the Environment,  
Ecological Design & Sustainability

SMITHIES IN  
SDS

# SUMMER SHARE-OUT



lunch provided for  
the first 25 people

**Tuesday, September 22nd**

**12:15pm**

**Sabin-Reed 301**

Share about your summer or just  
come to hear about your peers and  
professors! Send a google slide  
with photos to

***smithiesinsds@smith.edu***

*(doesn't have to be sds related!)*



join the  
SSDS club  
Discord!



# Plan for Today

- Recap last class
- What is perception?
- Pre-attentive processing
- Using perception to rank visual channels
- Takeaways

Note: I'm going to flash a bunch of slides quickly today. If that doesn't work for you, you are welcome to step out, close your eyes, etc.. (none of us will be offended and I will fill you in on anything crucial you missed!)

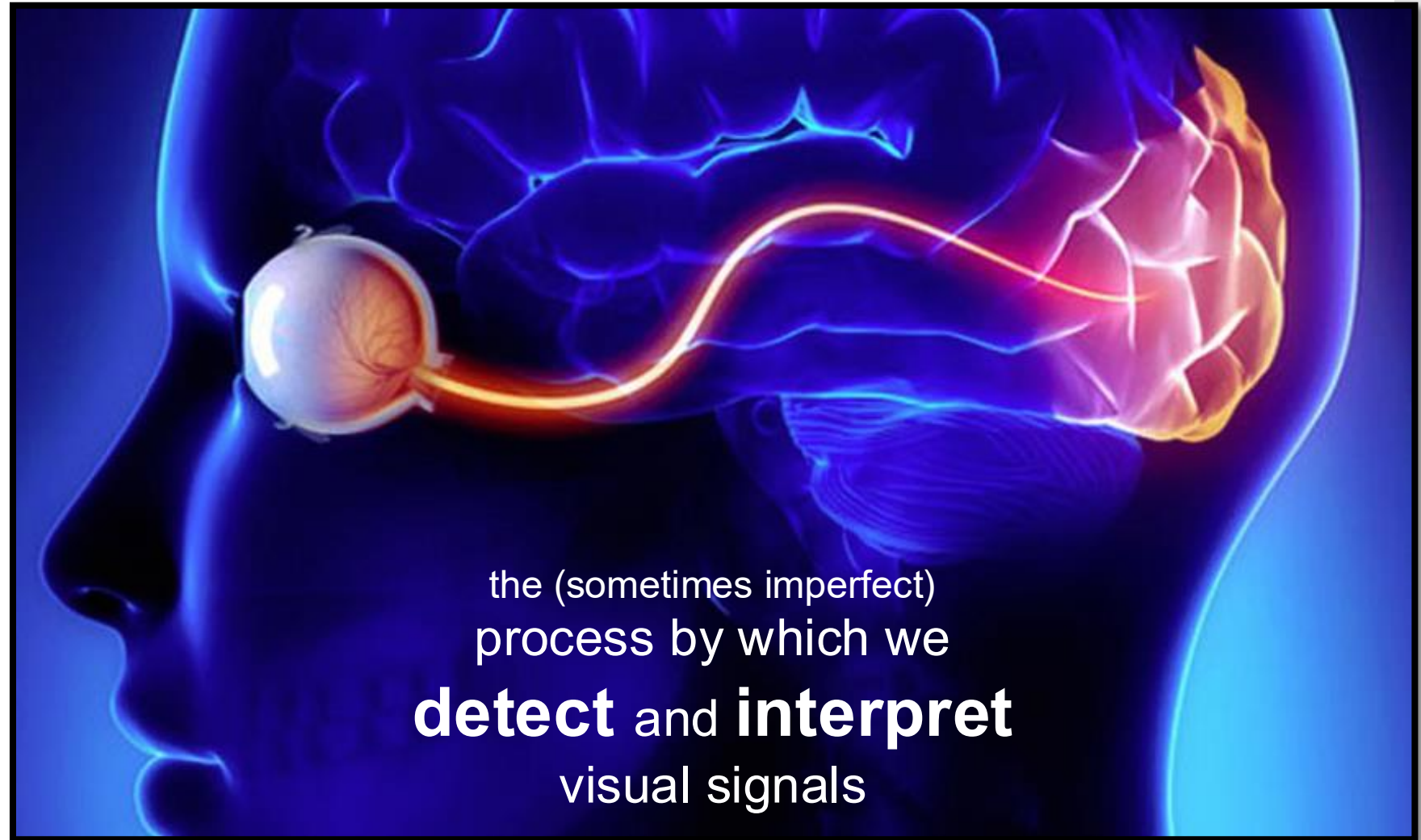
# Recap

- Work in small groups
- Write a checklist guide to decomposing a visual encoding (we've been using the same 4 steps in our examples, what are they?)

What is  
perception?

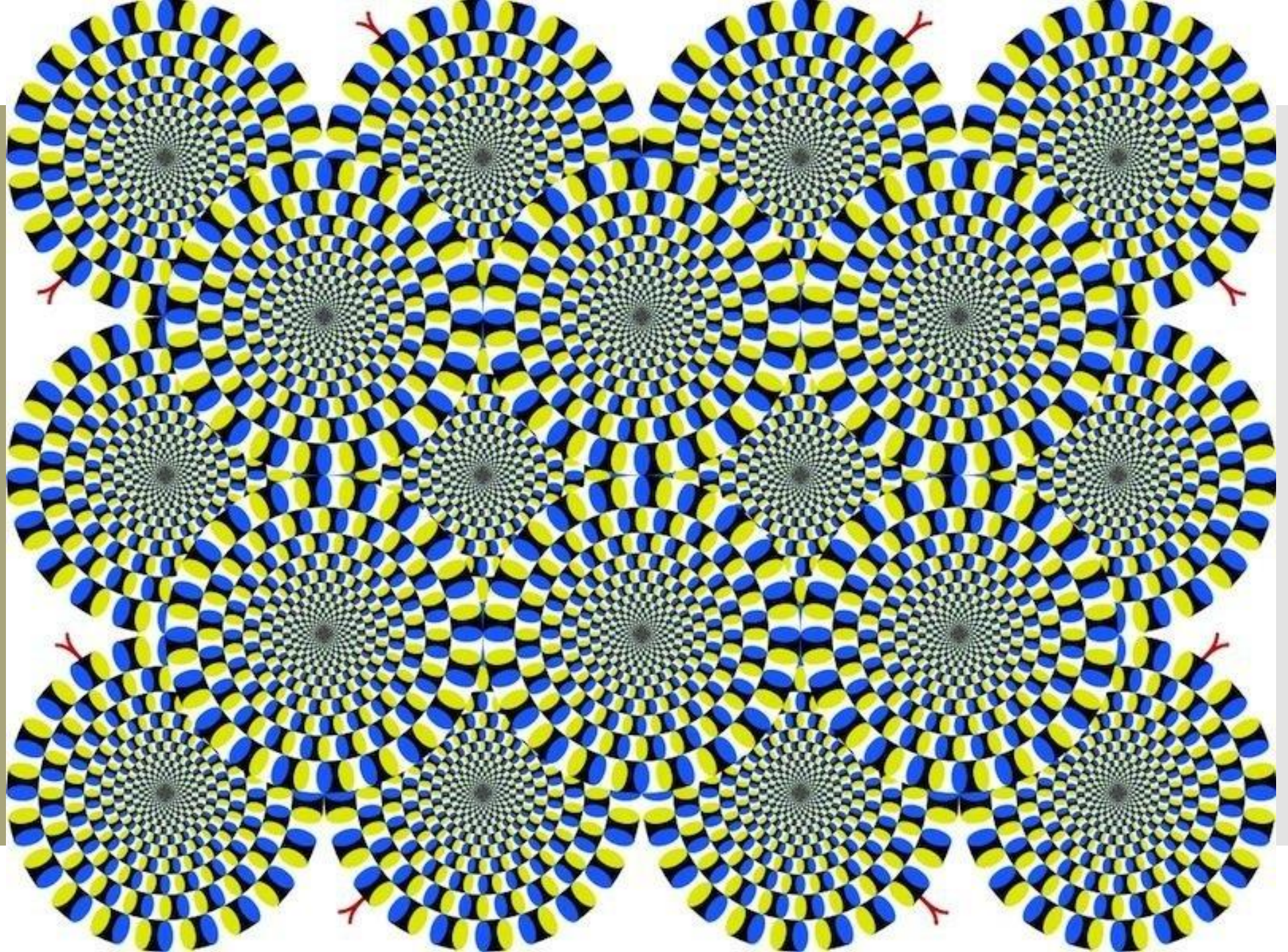


# Visual perception (def.)

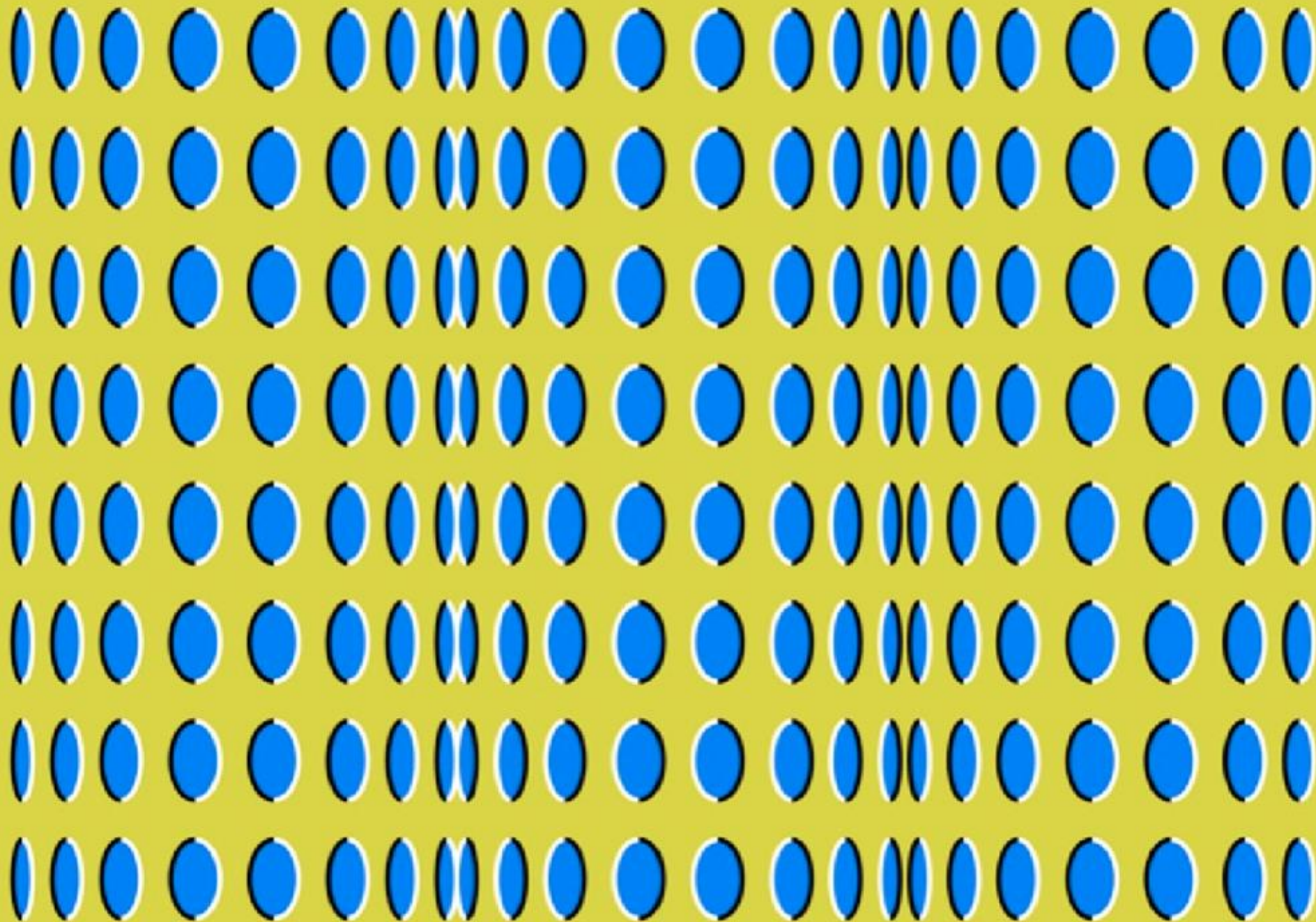


the (sometimes imperfect)  
process by which we  
**detect** and **interpret**  
visual signals

Do you see  
movement?

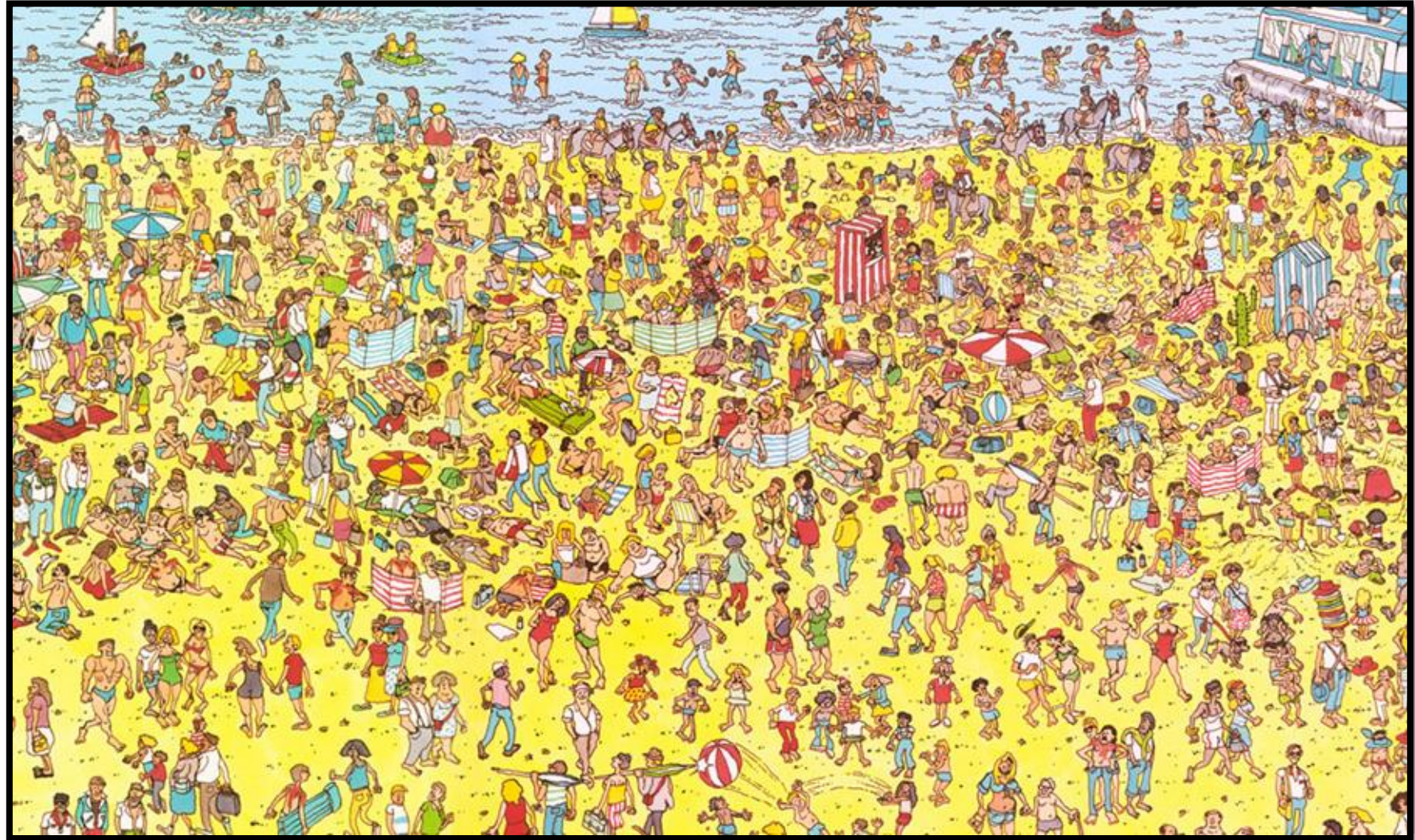


Do you see  
movement?



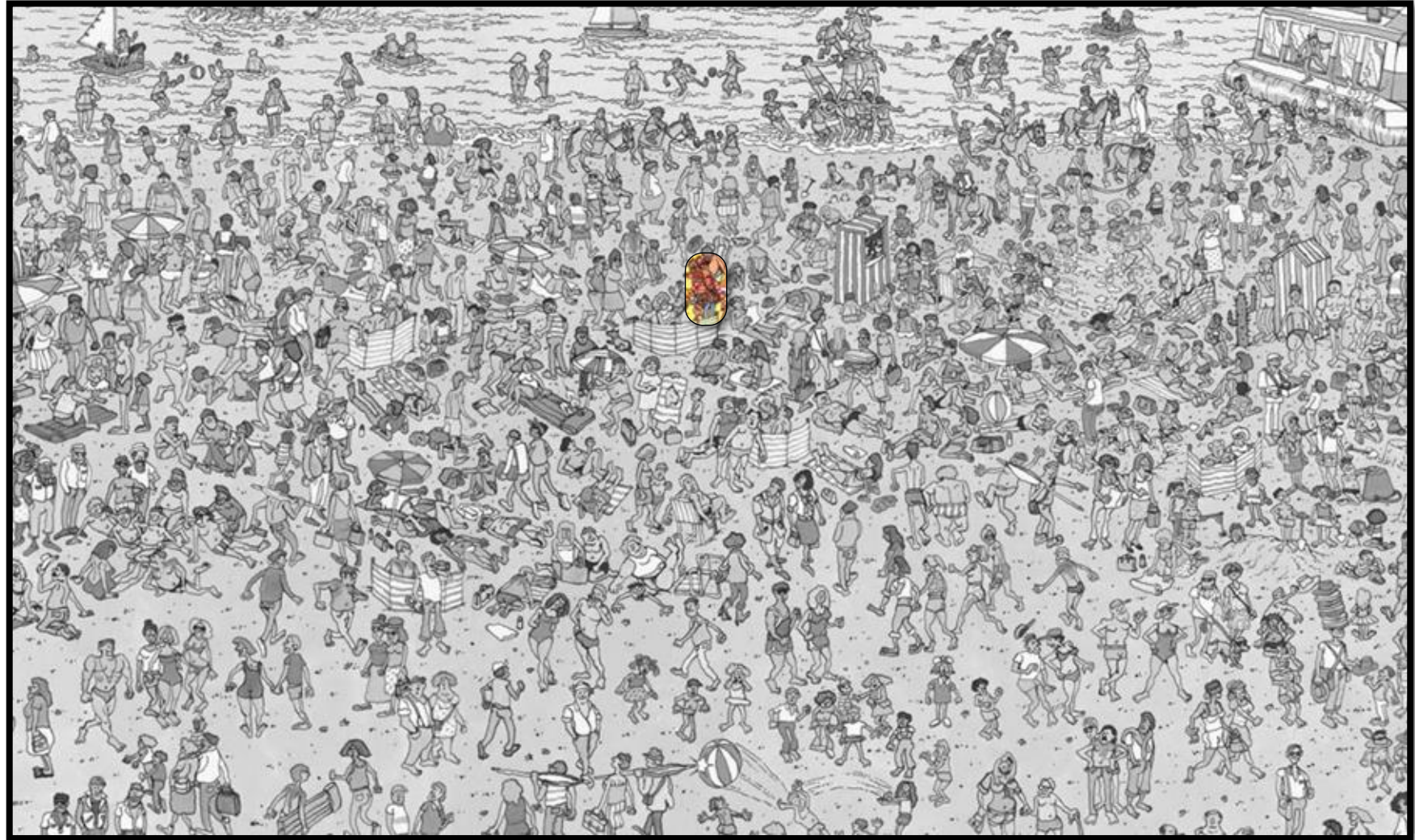
# Visual perception

Some things are processed slowly



Others are incredibly fast

Visual  
perception



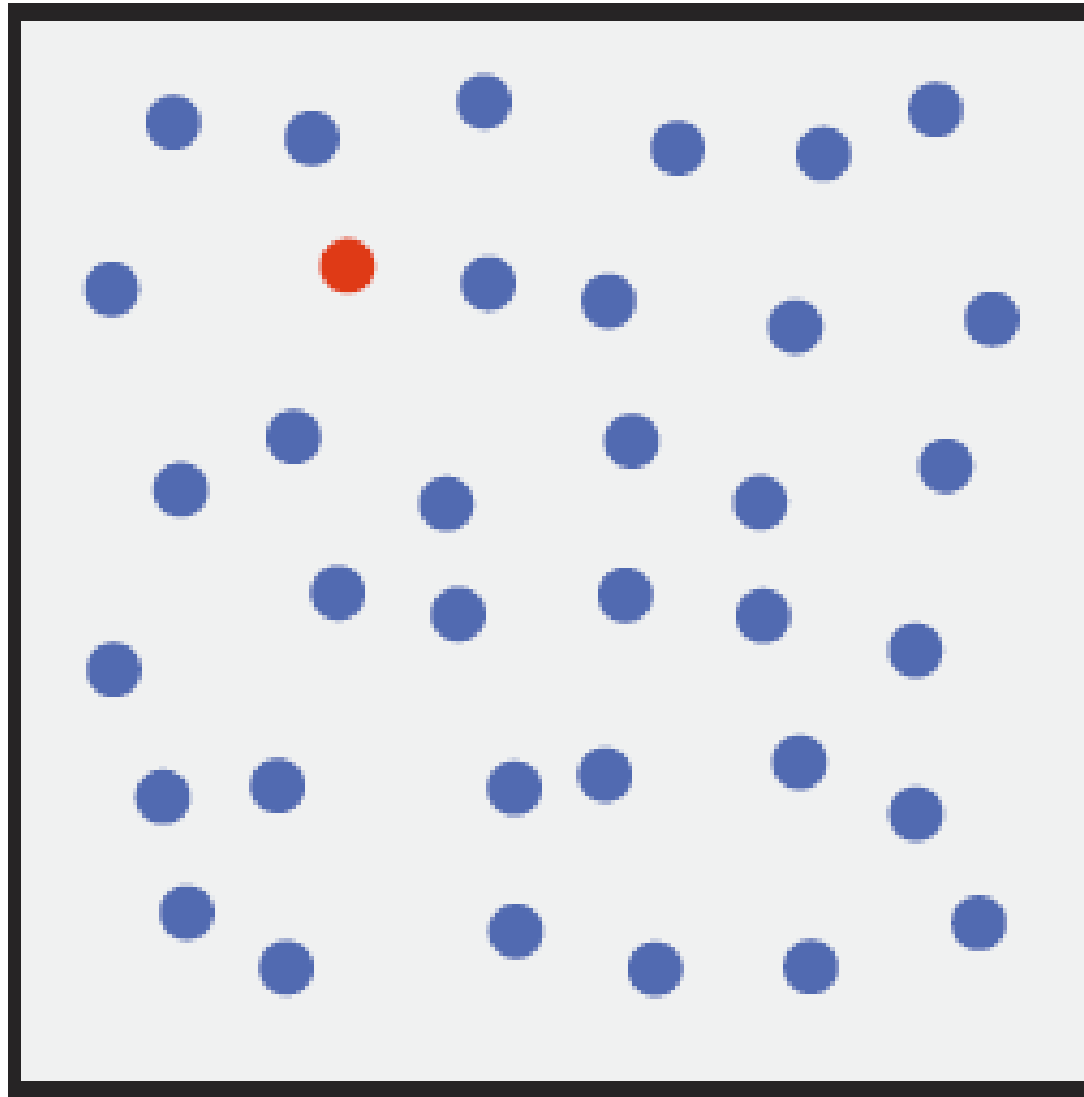
# Pre-attentive processing

Fast = “pre-attentive processing”

- Things that happen in <200ms of visual stimulation
- Performed in parallel across the entire visual field
- Example...

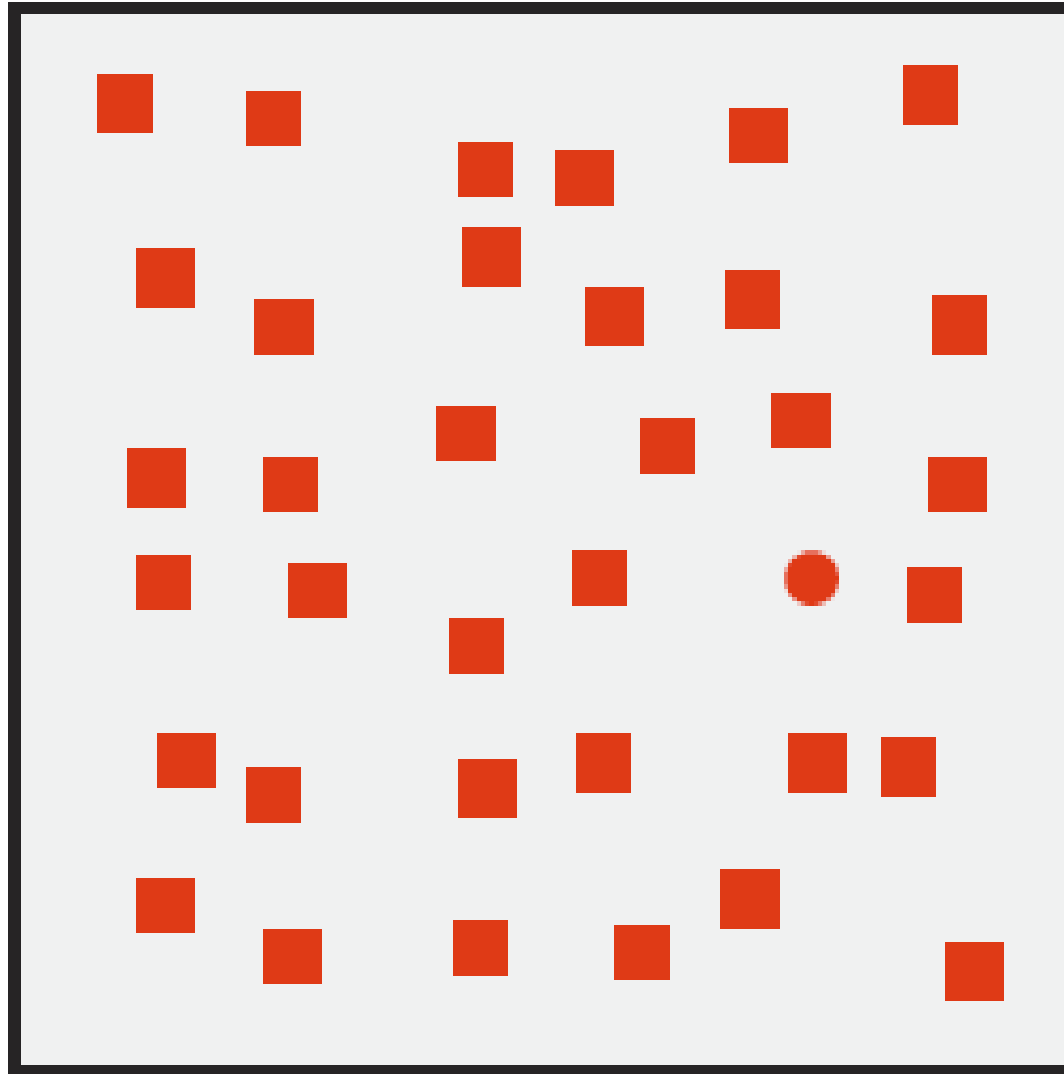
C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Pre-attentive processing



C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Pre-attentive processing



C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Pre-attentive processing

Fast = “pre-attentive processing”

- Things that happen in <200ms of visual stimulation
- Performed in parallel across the entire visual field
- A target with a unique visual property (ex. hue, curvature) “pops out” of a display; it can be easily detected

C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Pre-attentive processing

How do you think we as visualization designers can leverage pre-attentive processing?

C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

## Pre-attentive processing

On the next slide count how many zeros you see as fast as you can. Raise your hand (do not shout the number) when you have the answer.

## Attentive counting

1281768756138976546984506985624982826762  
9809858458224519856458945098459985943585  
9891330249945959595772564675750678974567  
8845789819821677654876364968560912949686

**How many zeros are there?**

Attentive  
counting

**We'll do the same on the next slide for threes.**

## Pre-attentive counting

12817687561**3**8976546984506985604982826762  
980985845822450985645894509845098094**3**585  
90910**3**0209905959595772564675050678904567  
8845789809821677654876**3**64908560912949686

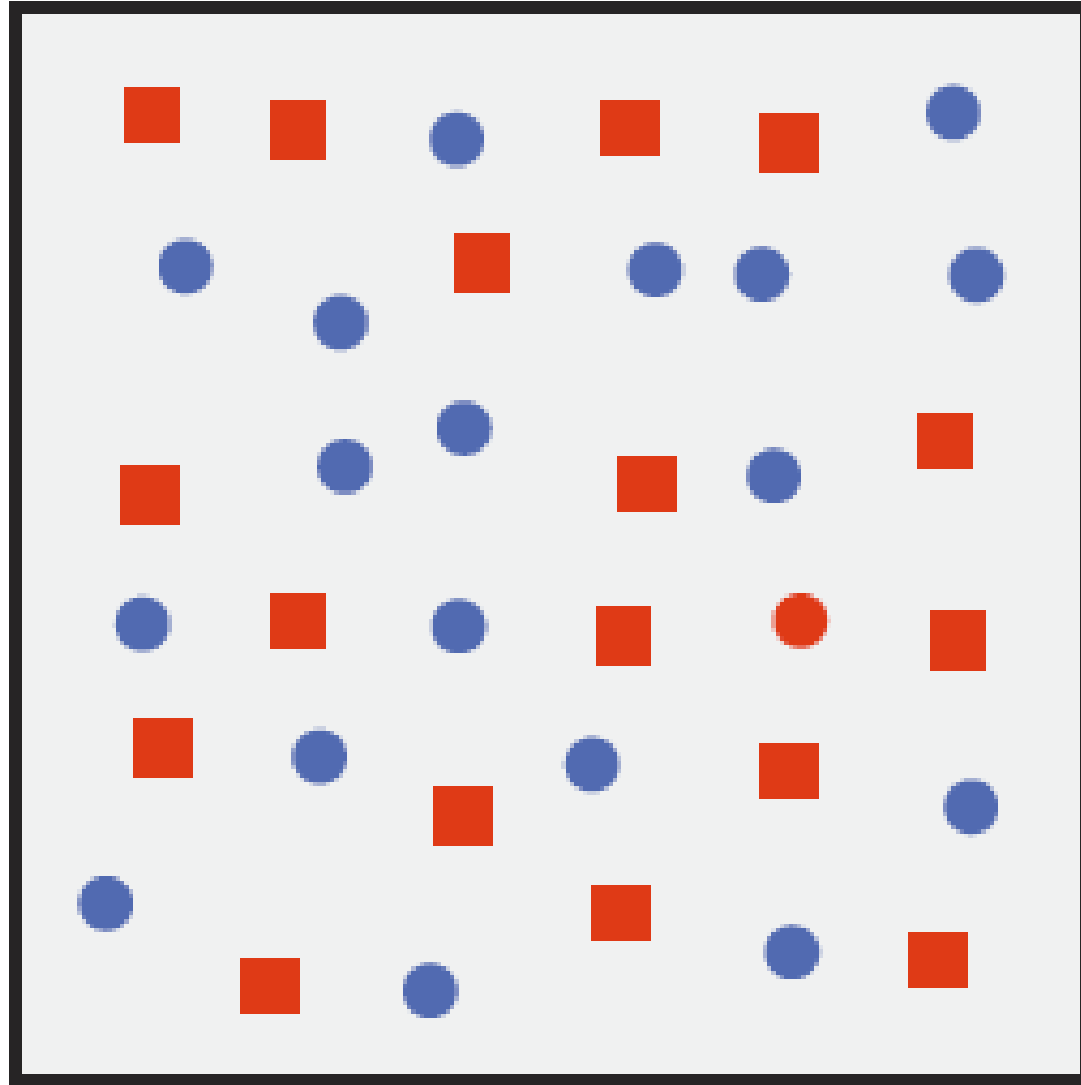
**How many threes are there?**

# Pre-attentive processing

Can we combine pre-attentive visual features to make something “pop out” even more?

C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

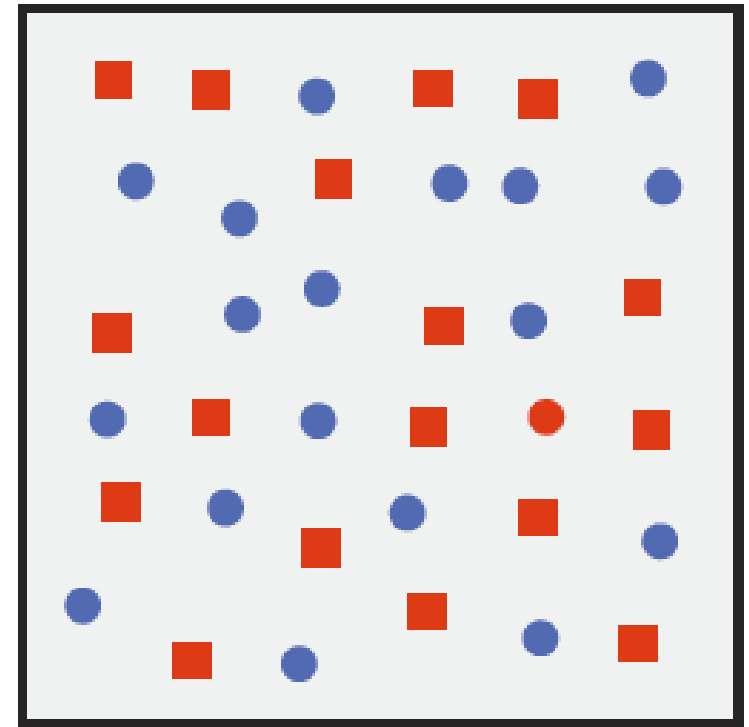
# Pre-attentive processing



C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

## Pre-attentive processing

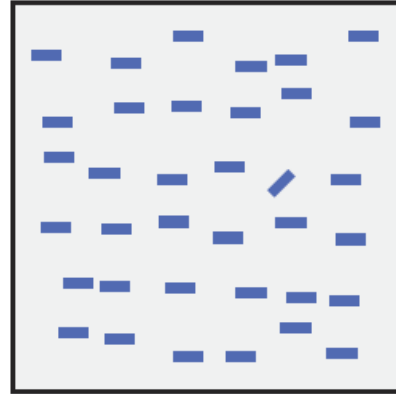
- A conjunction search (multiple visual features that are each present in distractors) is not pre-attentive
- Hue returns squares and the red circle
- Curvature returns blue circles and the red circle
- In both cases, you have to search the set returned



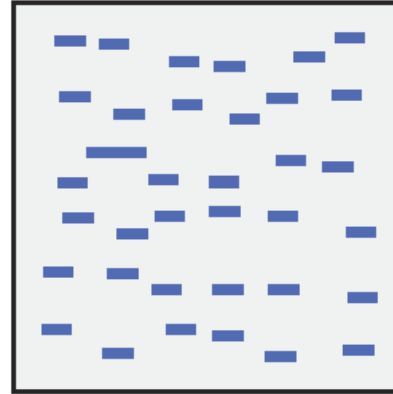
C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Pre-attentive processing

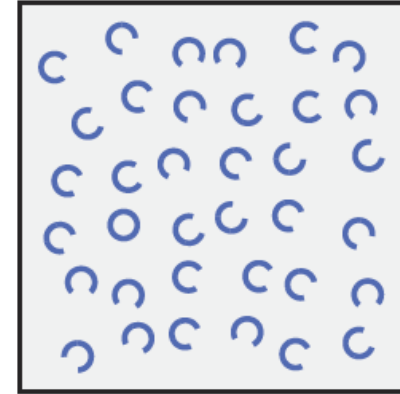
- Some pre-attentive features



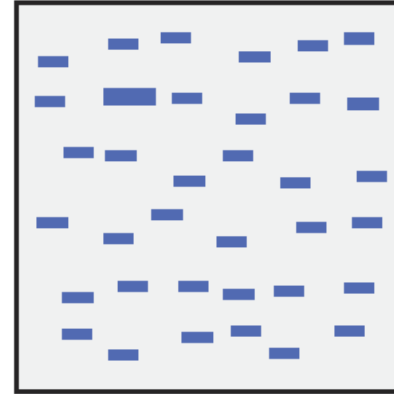
orientation  
[16], [17], [18], [19]



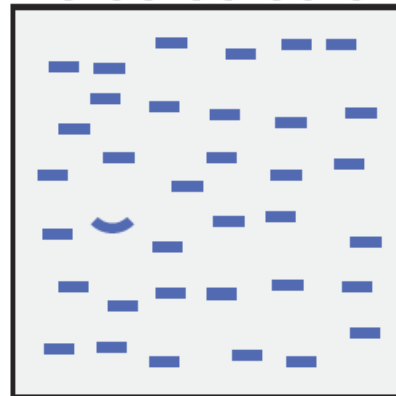
length  
[20], [21]



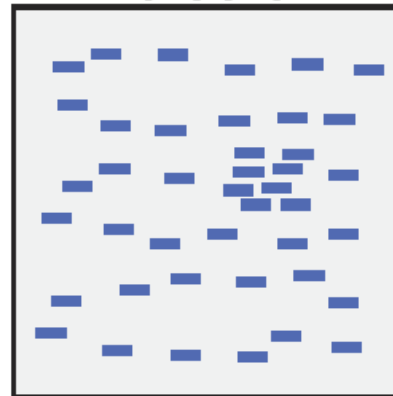
closure  
[16]



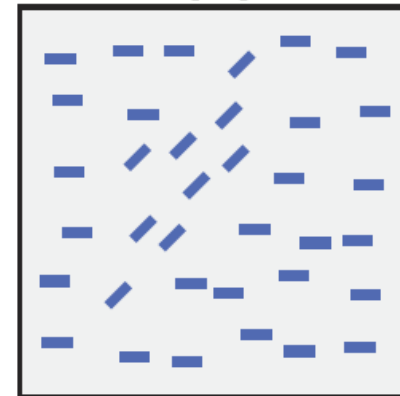
size  
[22], [23]



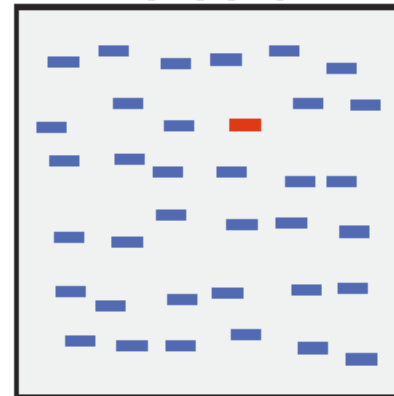
curvature  
[21]



density  
[23]



number  
[20], [24], [25]

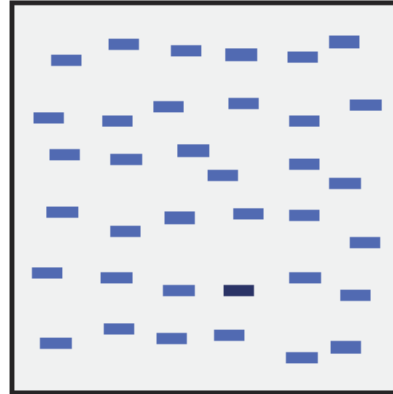


hue  
[23], [26], [27], [28], [29]

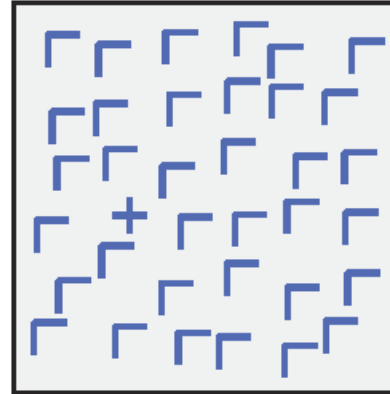
C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Pre-attentive processing

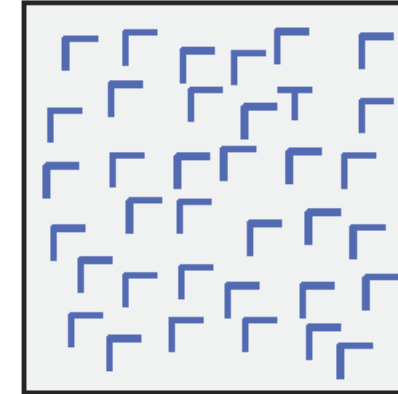
- Some pre-attentive features



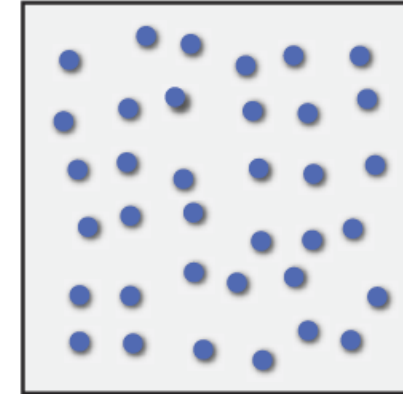
luminance  
[21], [30], [31]



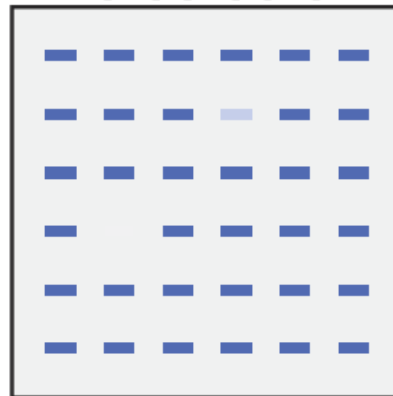
intersections  
[16]



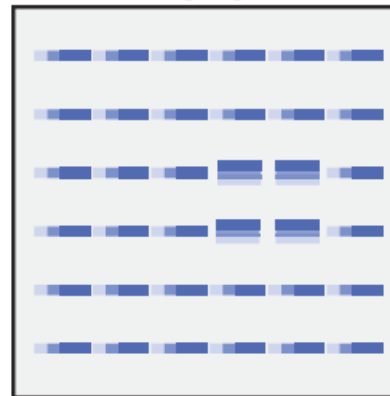
terminators  
[16]



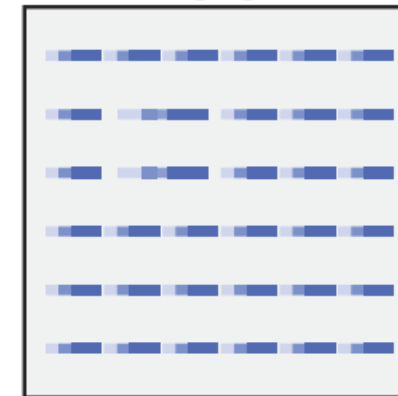
3D depth  
[32], [33]



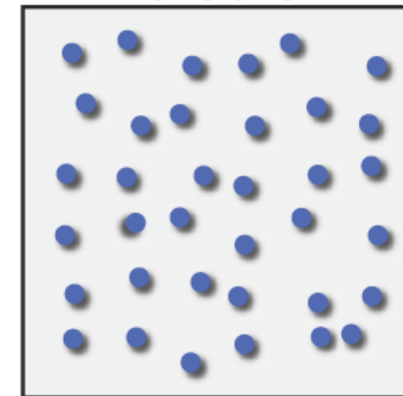
flicker  
[34], [35], [36], [37], [38]



direction of motion  
[33], [38], [39]



velocity of motion  
[33], [38], [39], [40], [41]



lighting direction  
[42], [43]

C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Pre-attentive processing

Pre-attentive features have been shown to support:

## **Target detection**

- Viewers detect a target element with a unique visual feature within a field of distractor elements

## **Boundary detection**

- Viewers detect a texture boundary between two groups of elements, where all of the elements in each group have a common visual property

## **Region tracking**

- Viewers track one or more elements with a unique visual feature as they move in time and space

## **Counting and estimation**

- Viewers count the number of elements with a unique visual feature

C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Visual Expectation and Memory

Pre-attention focuses on what draws the eye.

Visual memory asks, what do people remember from a glance?

C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Visual Expectation and Memory

Pre-attention focuses on what draws the eye.

Visual memory asks, what do people remember from a glance?

What are your hypotheses?

C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Visual Expectation and Memory

- Visual processing is a dynamic process.
- Your visual system is constantly building short-lived models of the external works that are specifically designed for your current visually guided tasks.
- There does not appear to be any general purpose vision. What we “see” when confronted with a new scene depends as much on our goals and expectations as it does on the light that enters our eyes.
- What is the implication here for visualization?

C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Visual Expectation and Memory

## Things to look out for: Change Blindness

- An interruption in what is being seen—a blink, an eye saccade, or a blank screen—renders us “blind” to significant changes that occur in the scene during the interruption.



C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

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# Visual Expectation and Memory

## Things to look out for: Inattentional Blindness

- Viewers can completely fail to perceive visually salient objects or activities when attending elsewhere
- <https://www.youtube.com/watch?v=vJG6g8U2Mvo>

C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Visual Expectation and Memory

## Things to look out for: Attentional Blink

- When two targets are presented in rapid succession, the second of the two targets cannot be detected or identified when it appears within approximately 100-500 msec following the first target
- <https://www.youtube.com/watch?v=kVvzpviOB7o>

C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

# Visual Expectation and Memory

What are the implications of Change blindness, Inattentional Blindness, and Attentional Blink for visualization?

C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127.

## Ranking Visual Channels

- Leveraging the perceptual system for visualization goes beyond understanding pre-attention and visual memory
- It is also a means for ranking visual channels.

# Recall

Flashback—

- What three types of data do we use in visualization?
- What two major categories do we use for visual channels?
- Name 7 visual channels.

# Ranking Visual Channels

- Using the correct visual channel type for the correct variable type is called **expressiveness**
  - Ex. A Magnitude channel, like luminance, for a quantitative or ordinal variable
  - Ex. An identity channel, like shape, for a nominal variable
- We also consider **effectiveness** to choose visual channels

# Ranking Visual Channels

- More important attributes should be encoded with more **effective** channels.
- **Effectiveness** refers to how well a channel supports:
  - Accuracy
  - Discriminability
  - Separability
  - Visual popout
  - Grouping

# Accuracy

**Definition:** how close human perceptual judgement is to an objective measurement of the stimulus

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**Definition:** how close human perceptual judgement is to an objective measurement of the stimulus



How much longer is the second bar?

# Accuracy

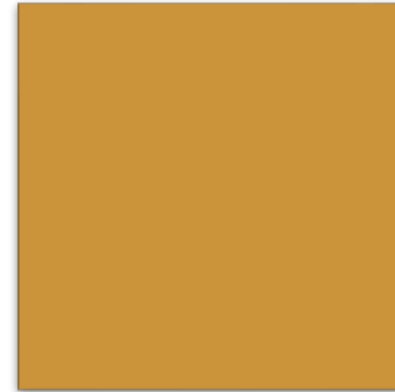
**Definition:** how close human perceptual judgement is to an objective measurement of the stimulus



How much longer is the second bar?  
2X

# Accuracy

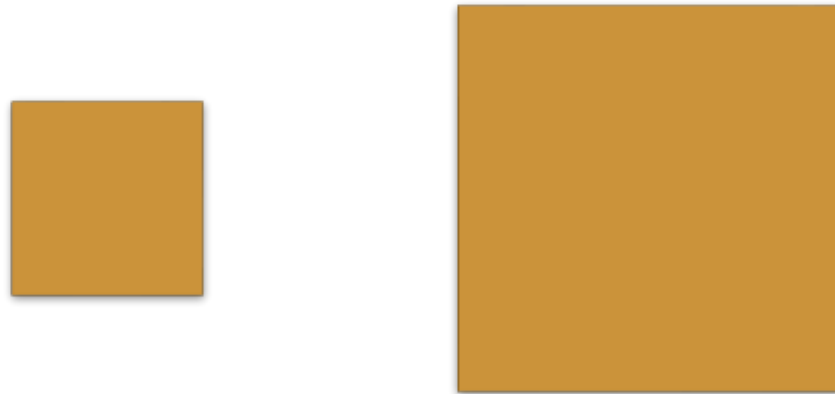
**Definition:** how close human perceptual judgement is to an objective measurement of the stimulus



How much bigger is the second square?

# Accuracy

**Definition:** how close human perceptual judgement is to an objective measurement of the stimulus



How much bigger is the second square?

4X

# Accuracy

**Definition:** how close human perceptual judgement is to an objective measurement of the stimulus



How much bigger is the second box?

# Accuracy

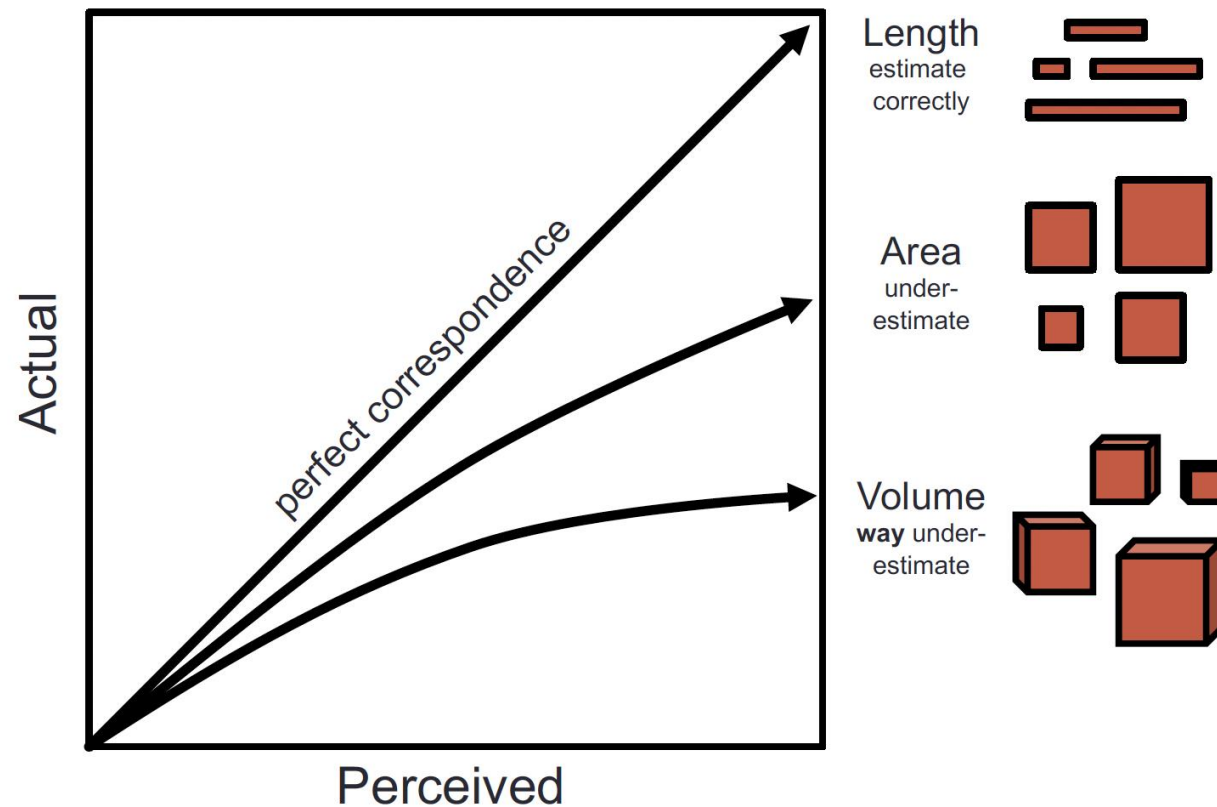
**Definition:** how close human perceptual judgement is to an objective measurement of the stimulus



How much bigger is the second box?  
27X

# Accuracy

**Definition:** how close human perceptual judgement is to an objective measurement of the stimulus

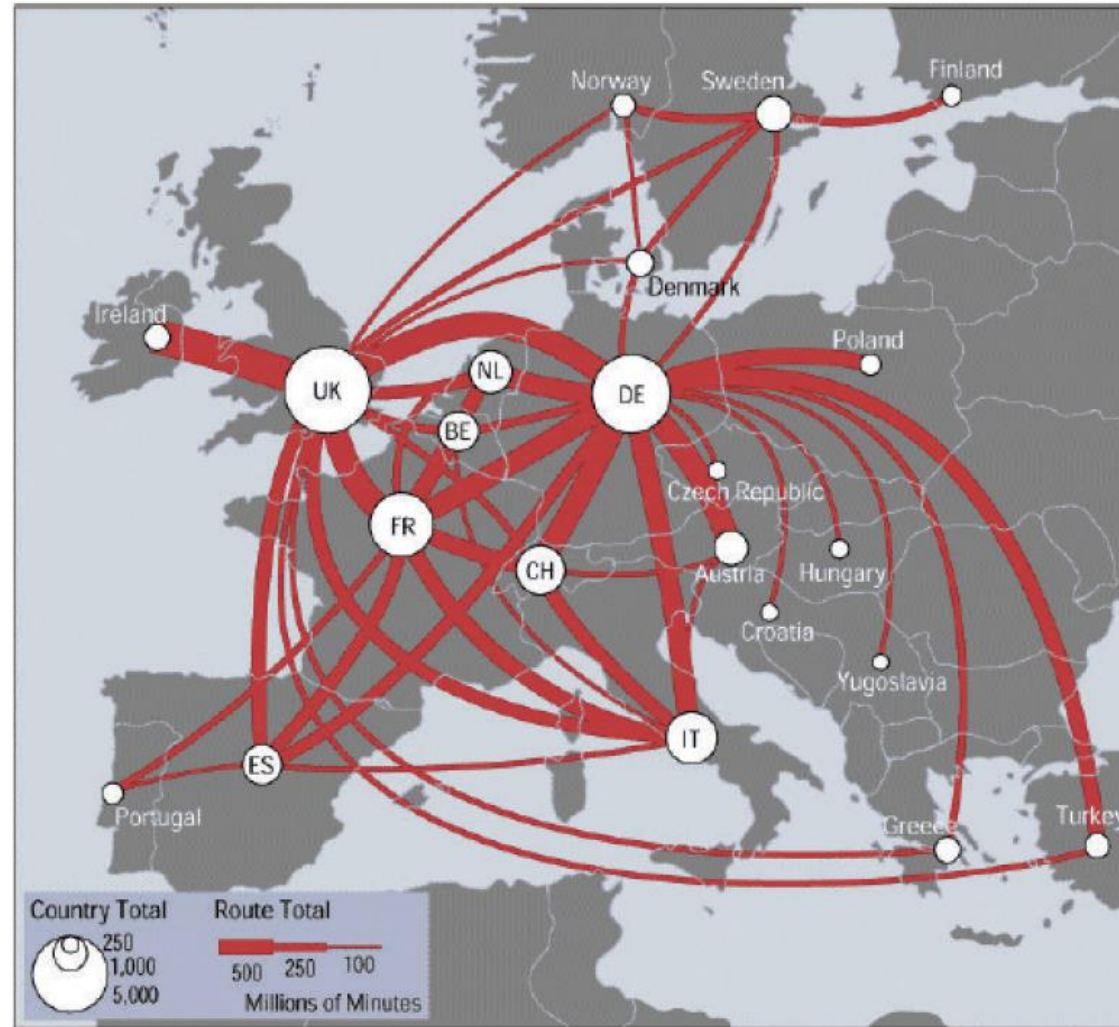


# Accuracy

Will a 2D or 3D bar chart result in better perceptual judgements? Why?

# Discriminability

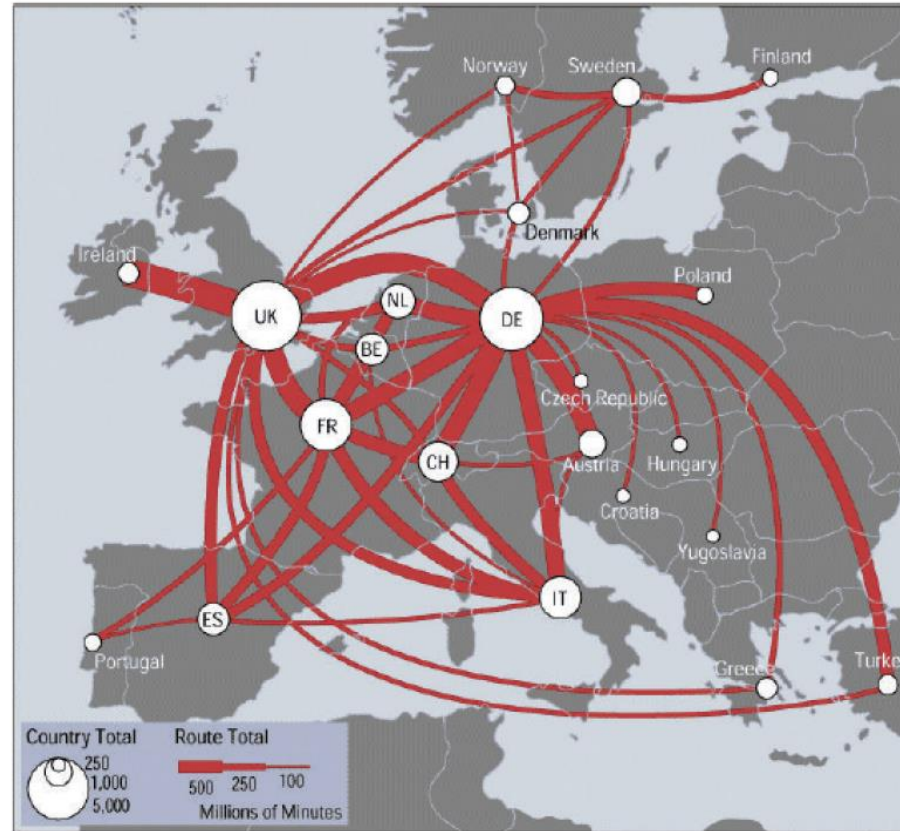
**Definition:** how differentiable levels of the channel are



<https://web.cse.ohio-state.edu/~shen.94/Melbourne/Slides/TamaraChp5.pdf>

# Discriminability

Could a different visual encoding make differences more clear? What might you try?

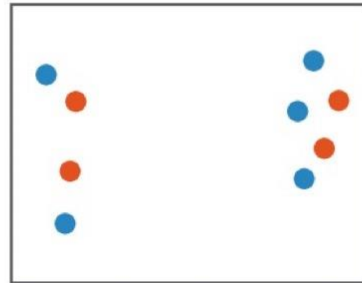


<https://web.cse.ohio-state.edu/~shen.94/Melbourne/Slides/TamaraChp5.pdf>

# Separability

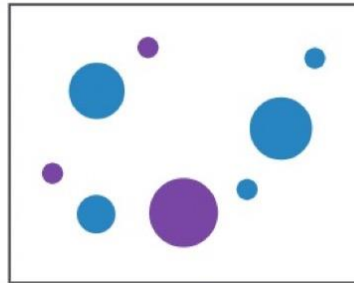
**Definition:** whether channels exist independently or integrally with others

Position  
+ Hue (Color)



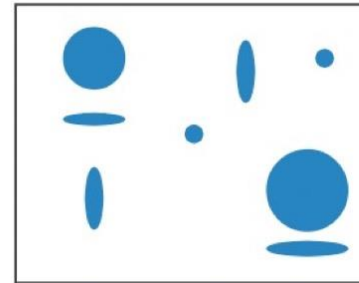
Fully separable

Size  
+ Hue (Color)



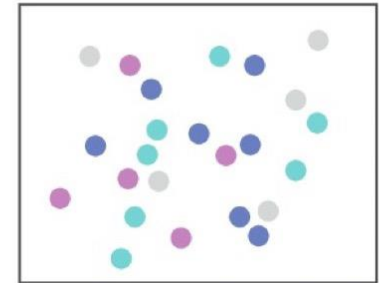
Some interference

Width  
+ Height



Some/significant  
interference

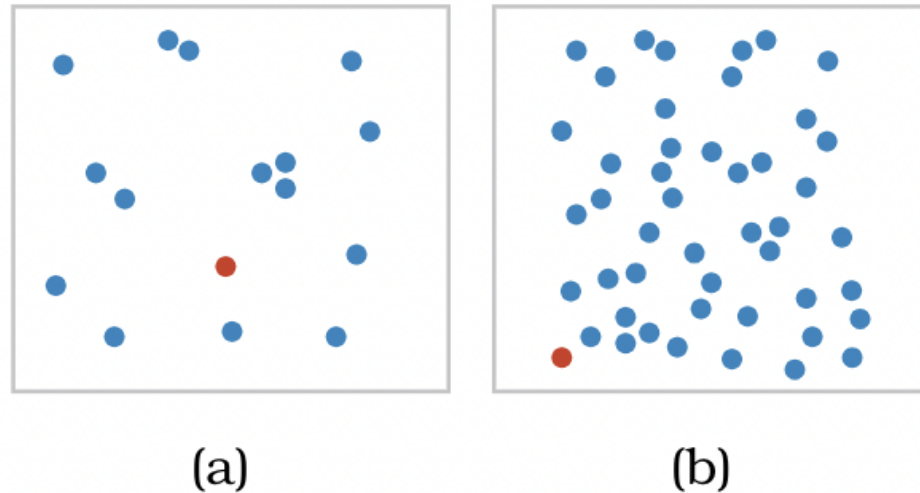
Red  
+ Green (saturation)



Major interference

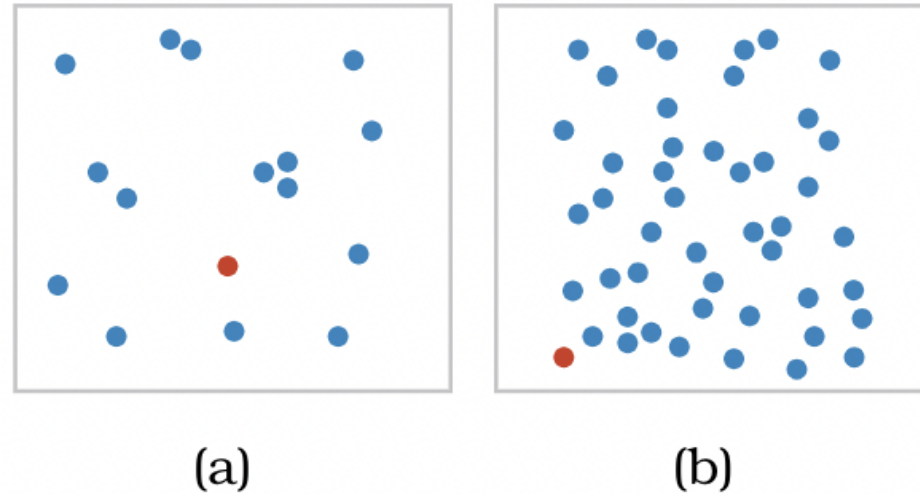
# Visual Popout

**Definition:** how well a distinct item stands out from others



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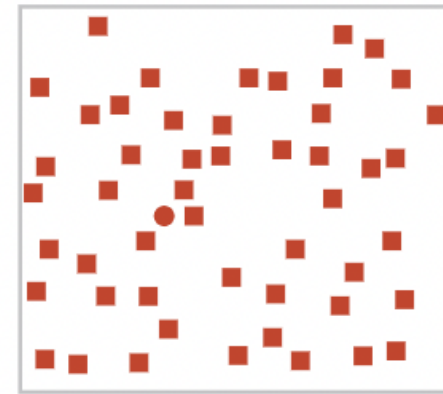
Color is a good channel for this

# Visual Popout

**Definition:** how well a distinct item stands out from others



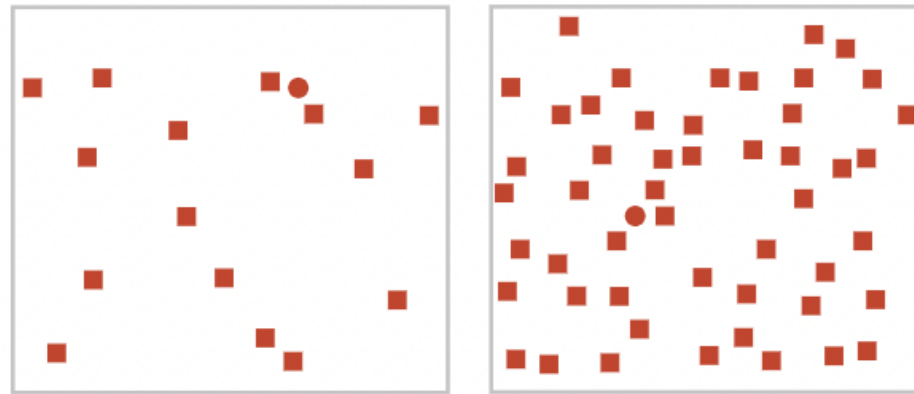
(c)



(d)

# Visual Popout

**Definition:** how well a distinct item stands out from others



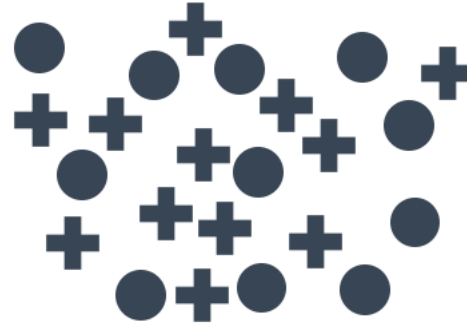
(c)

(d)

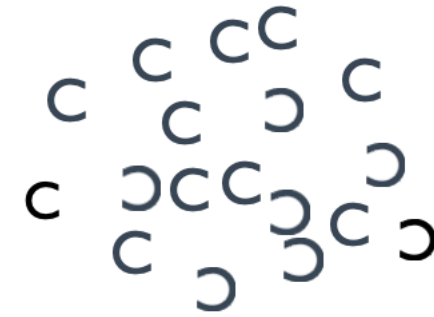
Shape is not as helpful

# Grouping

**Definition:** how likely people are to infer differences as representing distinct groups



Circles and +'s

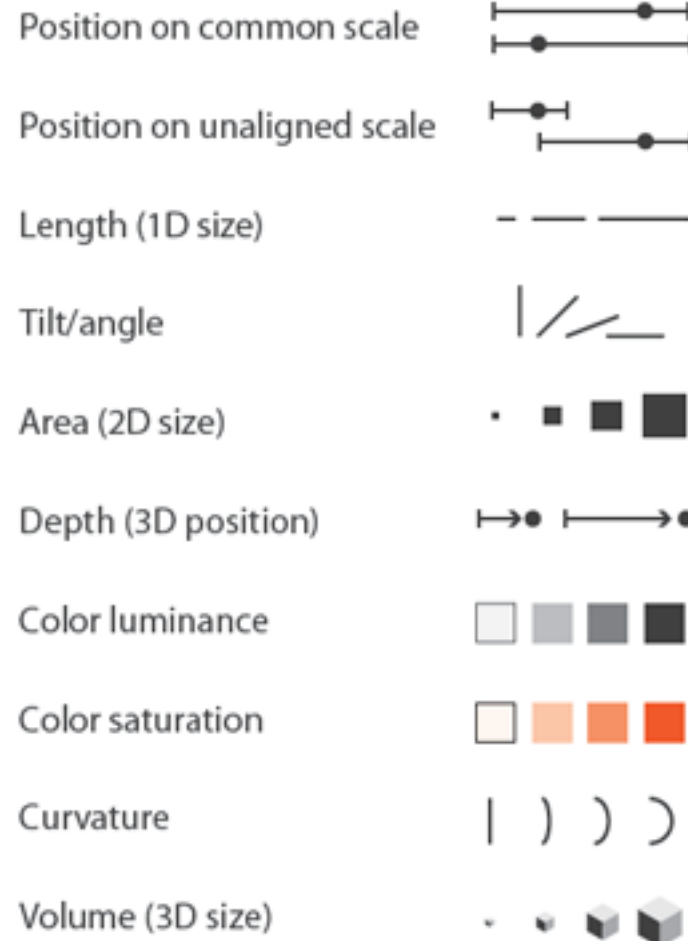


C's and D's

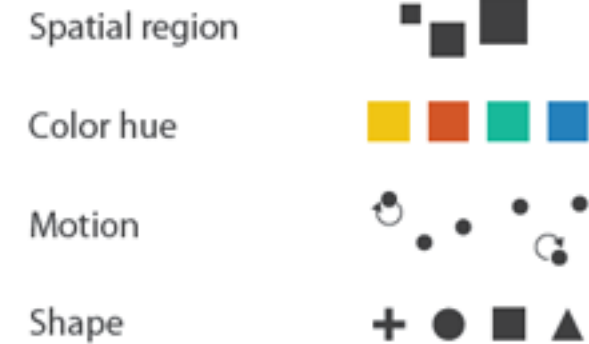
# Ranking Visual Channels

- **Design takeaway:** Prioritize most effective channels for most important data

## ➔ **Magnitude Channels: Ordered Attributes**



## ➔ **Identity Channels: Categorical Attributes**



Munzner, T. (2014). Visualization Analysis and Design (1st ed.). A K Peters/CRC Press. <https://doi.org/10.1201/b17511>

# The “gestalt effect”

**ge·stalt**

/gə 'SHtält,-'SHtôlt/

*noun* PSYCHOLOGY

an organized whole that is perceived as more than the sum of its parts.

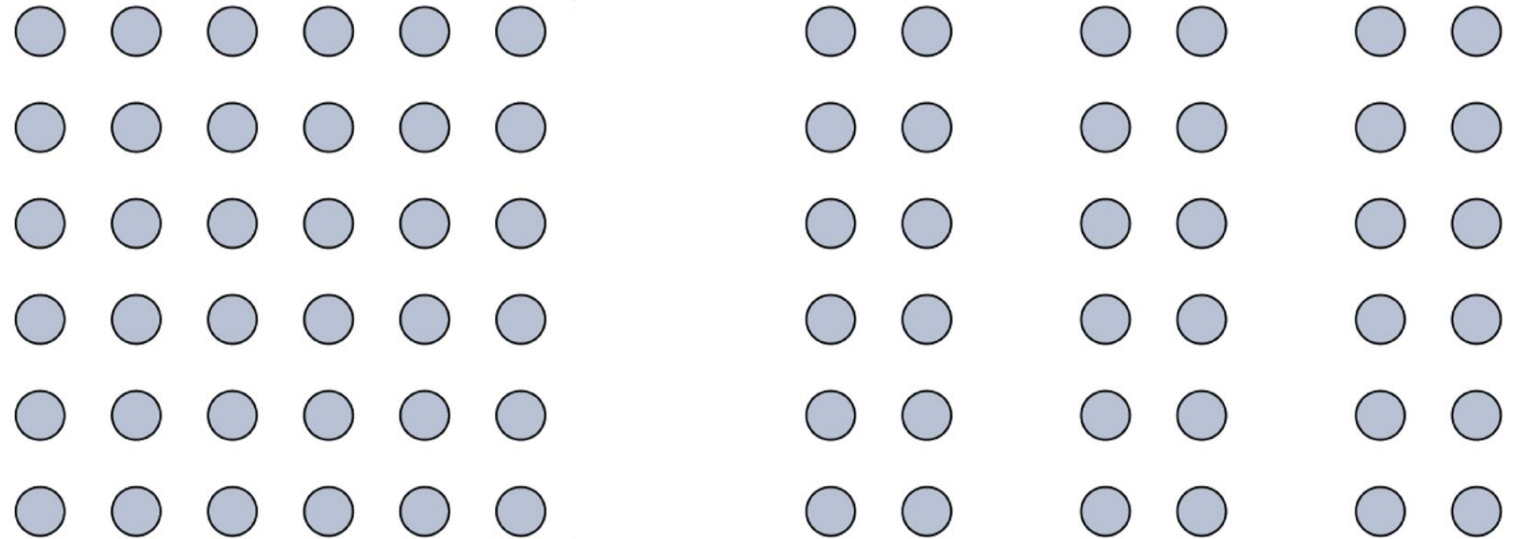


Translations, word origin, and more definitions

Our brain's ability to generate whole forms, instead of just collections of unrelated elements

# Law of Proximity

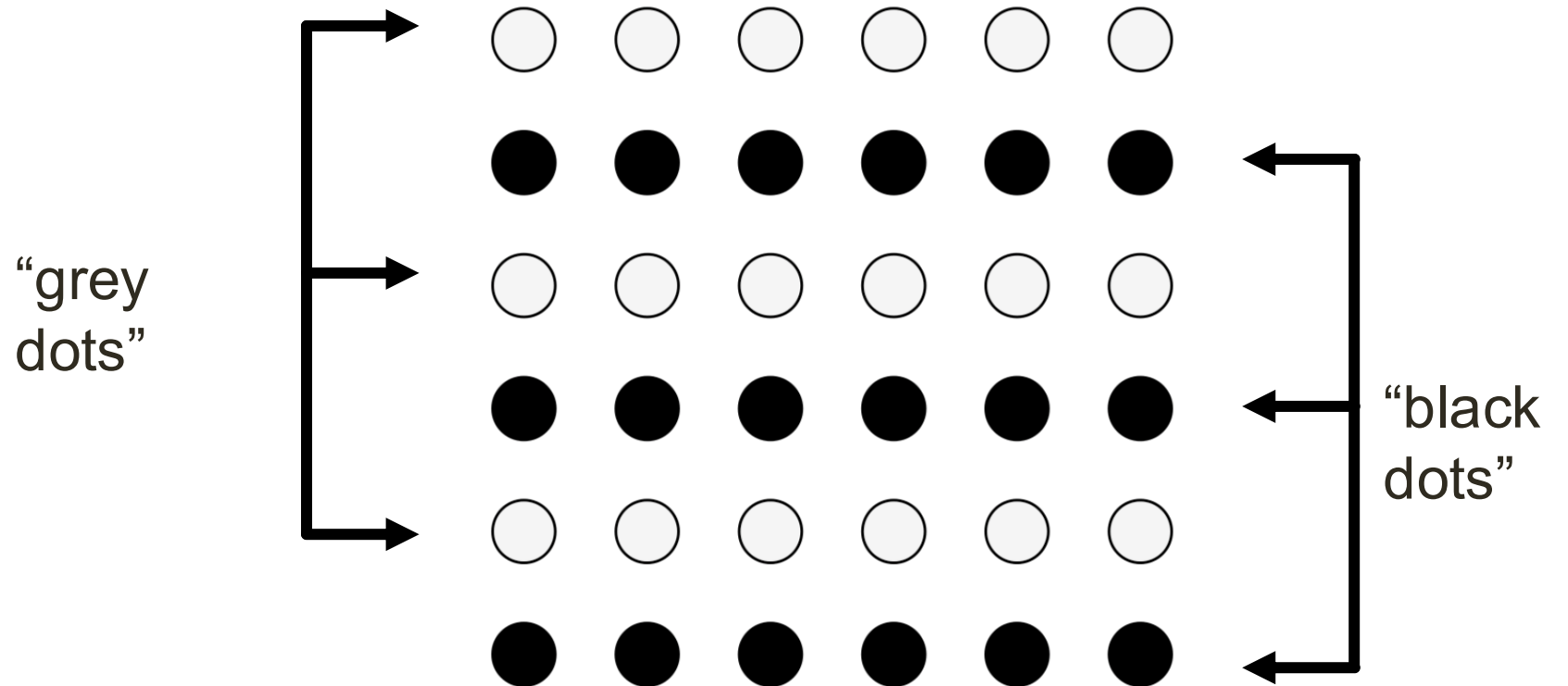
We interpret objects that are **close** to each other as a group



Wagemans, J., Elder, J. H., Kubovy, M., Palmer, S. E., Peterson, M. A., Singh, M., & von der Heydt, R. (2012). A century of Gestalt psychology in visual perception: I. Perceptual grouping and figure-ground organization. *Psychological bulletin*, 138(6), 1172–1217. <https://doi.org/10.1037/a0029333>

# Law of Similarity

We interpret objects that are **visually similar** to each other as a group



Wagemans, J., Elder, J. H., Kubovy, M., Palmer, S. E., Peterson, M. A., Singh, M., & von der Heydt, R. (2012). A century of Gestalt psychology in visual perception: I. Perceptual grouping and figure-ground organization. *Psychological bulletin*, 138(6), 1172–1217. <https://doi.org/10.1037/a0029333>

## Law of Closure

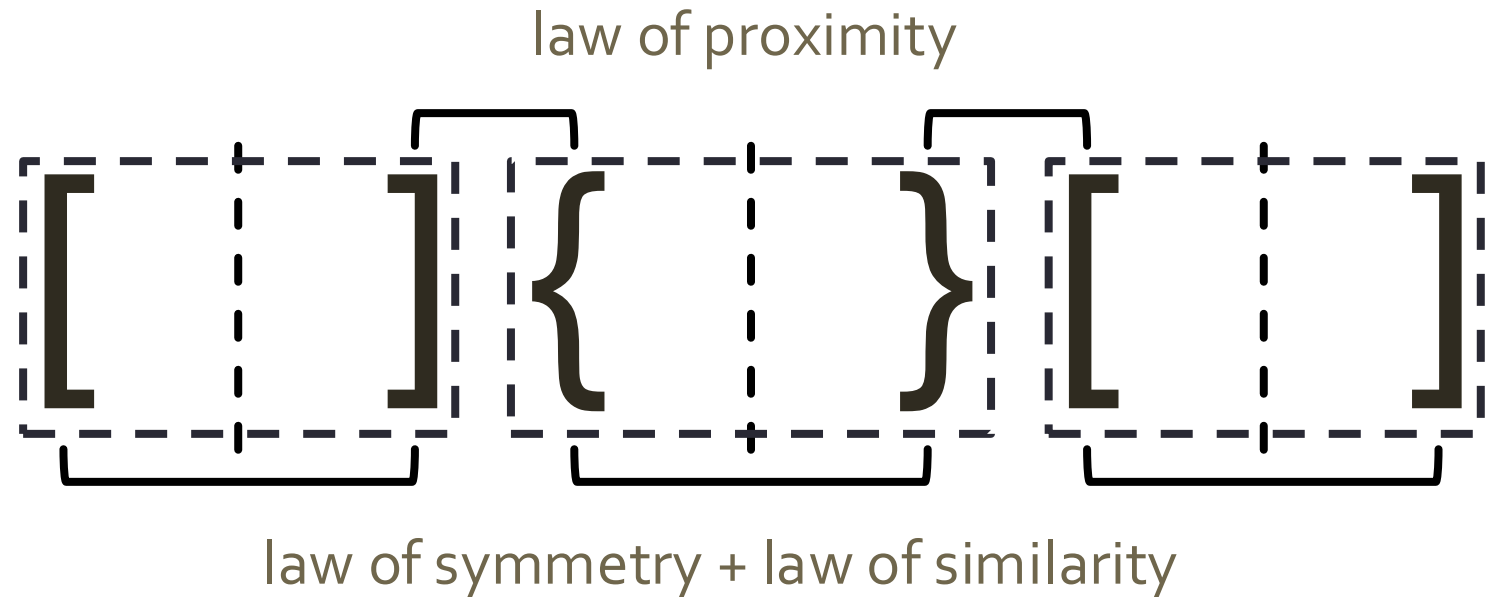
When parts of a picture are missing, we fill in the visual gap



Wagemans, J., Elder, J. H., Kubovy, M., Palmer, S. E., Peterson, M. A., Singh, M., & von der Heydt, R. (2012). A century of Gestalt psychology in visual perception: I. Perceptual grouping and figure-ground organization. *Psychological bulletin*, 138(6), 1172–1217. <https://doi.org/10.1037/a0029333>

# Law of Symmetry

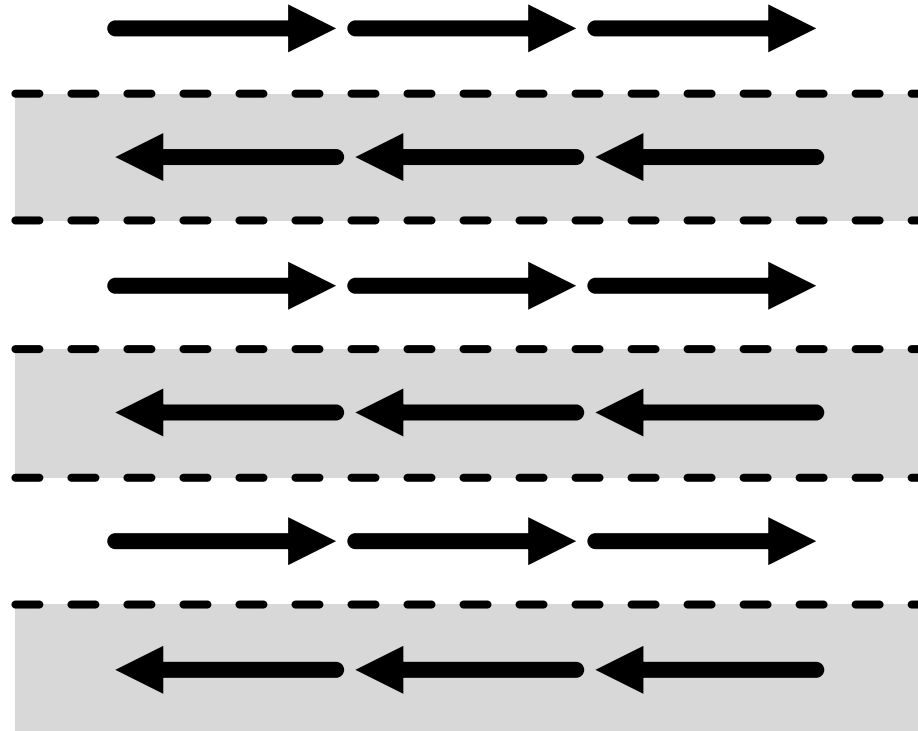
We perceive objects as being symmetrical, arranged around a center point



Wagemans, J., Elder, J. H., Kubovy, M., Palmer, S. E., Peterson, M. A., Singh, M., & von der Heydt, R. (2012). A century of Gestalt psychology in visual perception: I. Perceptual grouping and figure-ground organization. *Psychological bulletin*, 138(6), 1172–1217. <https://doi.org/10.1037/a0029333>

# Law of Common Fate

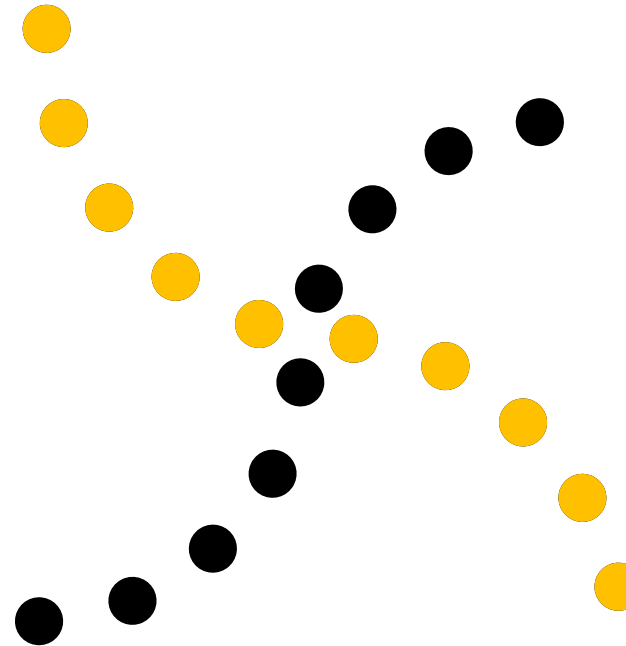
We group objects that we perceive to be moving along the same path



Wagemans, J., Elder, J. H., Kubovy, M., Palmer, S. E., Peterson, M. A., Singh, M., & von der Heydt, R. (2012). A century of Gestalt psychology in visual perception: I. Perceptual grouping and figure-ground organization. *Psychological bulletin*, 138(6), 1172–1217. <https://doi.org/10.1037/a0029333>

# Law of Continuity

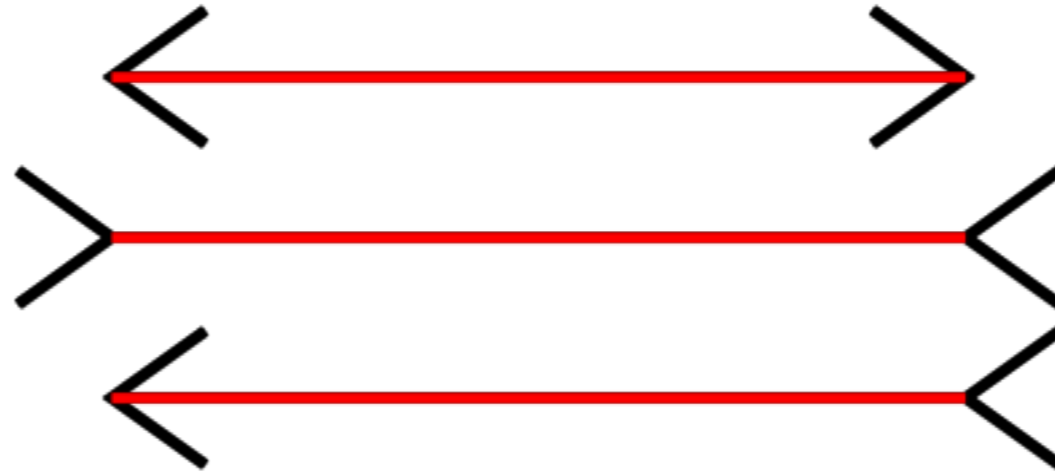
We tend to group objects along the **smoothest path**



Wagemans, J., Elder, J. H., Kubovy, M., Palmer, S. E., Peterson, M. A., Singh, M., & von der Heydt, R. (2012). A century of Gestalt psychology in visual perception: I. Perceptual grouping and figure-ground organization. *Psychological bulletin*, 138(6), 1172–1217. <https://doi.org/10.1037/a0029333>

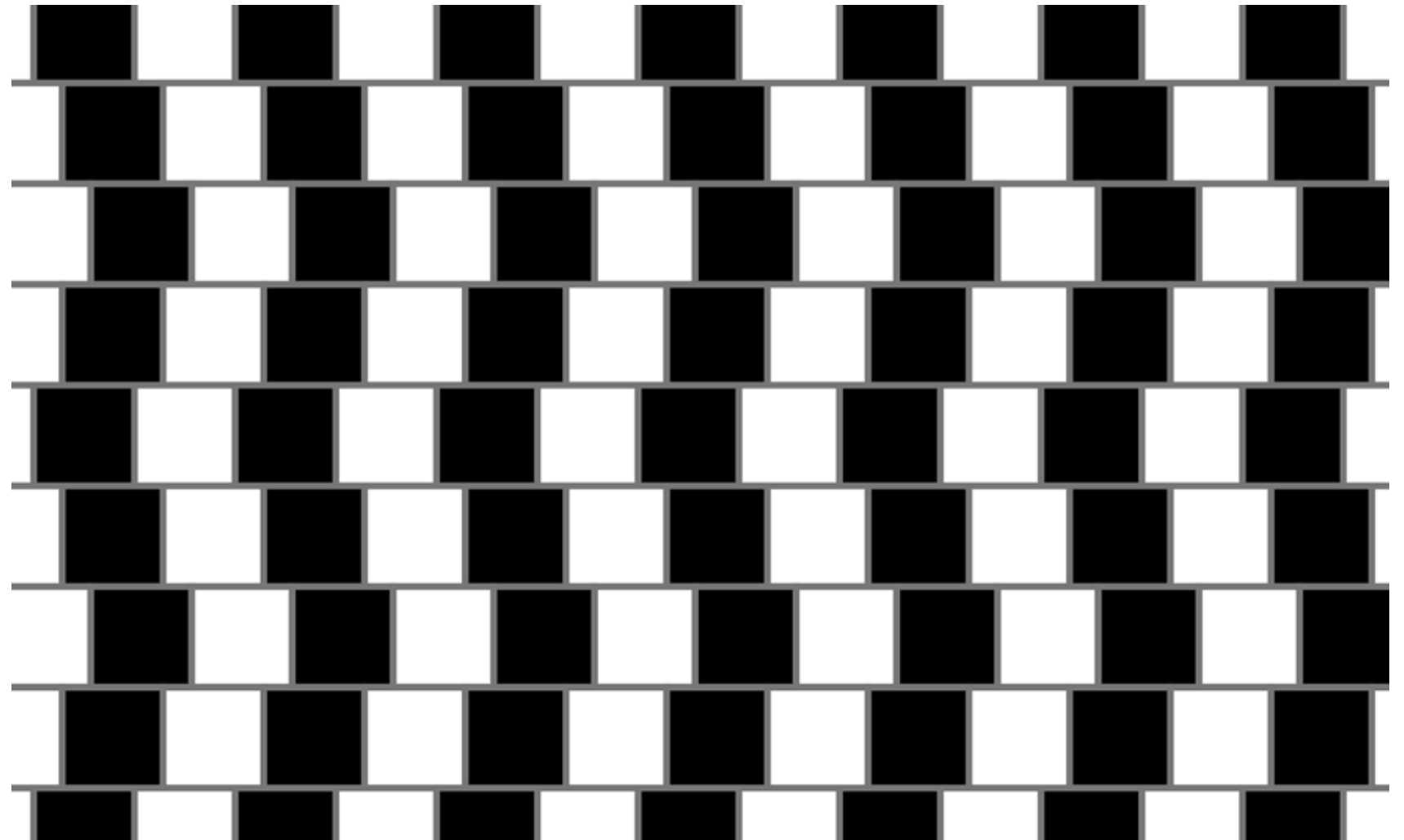
# Pre-attentive processing for visualization: The downsides

Sometimes gestalt & pre-attention compete



# Pre-attentive processing for visualization: The downsides

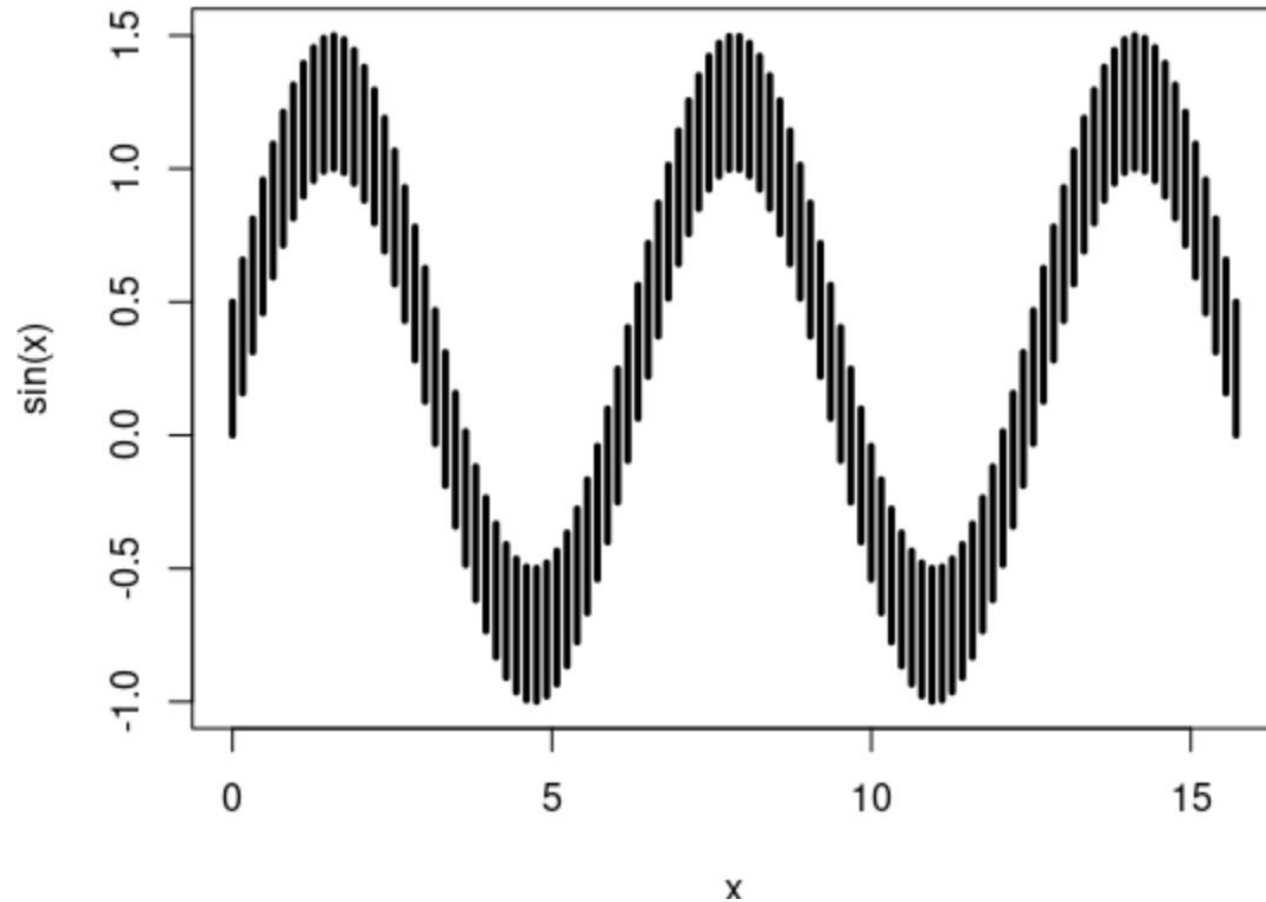
Sometimes gestalt & pre-attention compete



Gregory R. L. (1997). Knowledge in perception and illusion. *Philosophical transactions of the Royal Society of London. Series B, Biological sciences*, 352(1358), 1121–1127. <https://doi.org/10.1098/rstb.1997.0095>

# Pre-attentive processing for visualization: The downsides

Sometimes gestalt & pre-attention compete



Attentive  
processing

Short term “working” memory is limited

